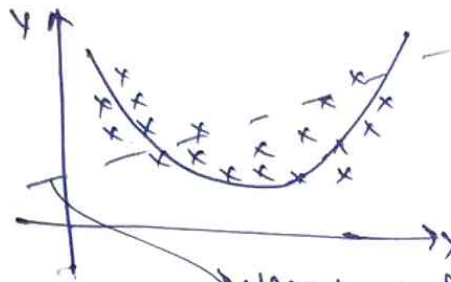


Polynomial Regression:



Very low R^2 score.

$$y = \beta_0 + \beta_1 x$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

(LR) (multiple)

degree $\rightarrow 2$
hyperparameter.

$$\begin{matrix} x_1 & y & (1,2) \\ \beta_5 & 100 & \\ \downarrow & & \\ x_1^0 & x_1^1 & x_1^2 & y \\ [1 & 35 & 35^2] & (1,4) \end{matrix}$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2$$

(for low degree underfit)

for high degree overfit)

$$x_1 \quad x_2 \quad | \quad y \quad \text{D} \quad \text{D=2}$$

$$y = \beta_0 + (\beta_1 x_1 + \beta_2 x_1^2) + (\beta_3 x_2 + \beta_4 x_2^2)$$

Polynomial Regression is a type of LR as in LR we check relationship b/w x and y ($\beta_1, \beta_2, \dots$)
As it is still linear, it is a type of LR